PROPOSAL DESCRIPTION, THEORETICAL BACKGROUND AND LITERATURE REVIEW:

General setting As critical, covariant superstrings have 10-dimensional target spaces, to be applicable to theories with 4-dimensional super-Poincaré invariance, these strings have to be "rolled up" on some manifold. There are many ways to do this which roughly fall into two categories to which one may briefly refer as "compactified" and "holographic". In the first case, the resulting model is often desired to resemble a "theory of everything", that is, a particle theory, perhaps of "grand unified" type, which includes the Standard Model of fundamental particles interacting with a theory of gravitation. The simplest manifolds achieving this goal are the famous Calabi-Yau 3-folds. In this particular type of manifold, the vanishing of a certain topological class guarantees that the original superstring continues to be exactly anomaly-free (critical).

Besides this important simplification, the Calabi-Yau condition implies the phenomenologically desirable condition of space-time supersymmetry: *If* the ultra-violet completion of the Standard Model is realized in Nature by a mechanism currently known to particle phenomenologists, there is ample reason to believe that it will employ this symmetry. Unfortunately, the simplest compactificatons are inadequate for phenomenological applications. Among the various difficulties is the "moduli problem": On general grounds, compactifications have scalar fields in their spectra and supersymmetric backgrounds require "flat directions" in their potentials which make them massless. These moduli can be lifted by turning on flux, that is, Maxwell-like field strengths supported on homology cycles in the internal space. In one of the two classes available for this line of attack, this involves the so-called "Ramond-Ramond" flux of type-II superstrings [1].

In the second type of theory, the resulting non-compact space may have a similar gravity+matter structure but now propagating on a background with anti-de-Sitter (adS) geometry. Such geometries have a boundary which can be reached in finite time by null geodesics in the interior (bulk). Boundary conditions for the bulk fields must be specified to complement the field equations. The constraints on these boundary conditions can be interpreted as conditions on the spectrum of observables of an auxiliary quantum gauge field theory (CFT) with certain symmetry properties – most notably, non-anomalous conformal invariance [2]. An important implication of this "adS/CFT correspondence" is a prescription relating the spectrum and correlation functions of the boundary gauge theory at strong coupling to the action of certain weak-field solutions to the bulk equations of motion [3]. Due to recent advances in this field of research, it is now believed that we may be on the verge of solving, for the first time, a non-trivial conformal field theory in four dimensions, namely maximally supersymmetric Yang-Mills theory in the 't Hooft limit [4].

In the simplest cases, backgrounds of this type result (in a certain "near-horizon" limit) from the gravitational back-reaction of a stack of coincident D-branes – the stringy source of the Ramond-Ramond fields described above. Therefore, the realization of this program again relies on the propagation of superstrings in a background of such fields.

Covariant superstrings Arguably, most interesting applications of string theory to the problems of theoretical physics are set in background space-times which include Ramond-Ramond flux. In turn, the understanding of superstrings in Ramond-Ramond backgrounds beyond the supergravity approximation requires a covariant description of the superstring.

The formulation of such strings has progressed steadily since the early 1990s. The first more-or-less covariant superstring [5] was the first of the so-called "hybrid" superstrings. This general class is technically equivalent to the RNS superstring in that it is constructed from the variables

defining the latter by a complicated series of field redefinitions mixing the matter fields and super-reparameterization ghosts. This procedure results in a new set of worldsheet variables which fall into two sets. The first is formally identical to a subset of the RNS matter variables and the latter are variables of Green-Schwarz-type transforming covariantly under a d-dimensional Lorentz subgroup of the 10-dimensional target space Lorentz group. For example, in the first such string constructed, the set of worldsheet fields consisted of 4-dimensional Green-Schwarz fields (those related to 4-dimensional, $N = 1$ superspace) together with their momenta, 6 additional coordinates together with their RNS worldsheet spinor partners, and a single decoupled time-like chiral boson. These variables have no branch cuts in their operator products with the 4-dimensional supersymmetry generators. It follows that any vertex operator constructed from these ingredients is automatically GSO-projected. A key addition to the Green-Schwarz variables in this description are the Siegel constraints – worldsheet weight-1 spacetime spinors canonically conjugate to the odd superspace coordinates and commuting with the supersymmetry generators [6]. The introduction of this field into the action as the (covariant version of the) momentum to the odd variables breaks the Siegel symmetry of the Green-Schwarz part but gives canonical kinetic terms for the odd field. Additionally, the Siegel symmetry is replaced with an $N = 2$ worldsheet superconformal symmetry, the generators of which are constructed from these constraints.

Subsequently to the 4-dimensional, $N = 1$ hybrid string, new hybrid strings preserving various amounts of target space super-isometries were constructed. The first examples, in increasing symmetry, preserved more Poincaré invariance but still only 4 supercharges. This increased to 6 in the $U(5)$ formalism [7] and finally the 6-dimensional hybrid with 8 manifest supersymmetries [8]. In this last case, extra odd coordinates were introduced together with new Siegel-type constraints to gauge them away. The new constraints were imposed by the BRST method of introducing commuting spinor ghosts for the constraints and constructing an odd charge squaring to 0. Finally, a formulation with maximal spacetime super-Poincaré invariance was described by constructing a BRST-like charge out of the full set of 10-dimensional Siegel constraints and using an auxiliary bosonic spinor "ghost" satisfying a set of constraints identifying it as a pure spinor [9]. The resulting pure spinor superstring has various unusual features such as a BRST-like symmetry not arising from the gauge fixing of any easily identified geometric structure (see, however, [10]). (This should perhaps not be too surprising given that in passing from the 6-dimensional string to that in 10, the focus on the structure of the worldsheet symmetries shifted from the $N = 2$ superconformal algebra to the BRST-type symmetry.) Subsequent to its discovery, the properties of this pure spinor string have been studied in detail – at least in 10-dimensional backgrounds with maximal supersymmetry.

Compactified backgrounds In the backgrounds which preserve fewer than the maximal number of supersymmetries, the pure spinor description is rather poorly understood. A very important example of a background of this type is the compactification of the pure spinor superstring on a Calabi-Yau 3-fold. Already, a straightforward dimensional reduction of the heterotic pure spinor string sheds light on the role of the pure spinor variable in ensuring that the physical state conditions are related to space-time supersymmetry [11]. In the straightforward reduction, as usual in superspace, the spinors do not reduce nicely as tensors do and one obtains a complicated set of component-type field variables. On general grounds, however, the 10-dimensional pure spinor is expected to reduce to a formalism of hybrid-type in which the 4-dimensional space-time part is represented by covariant Green-Schwarz-Siegel variables while the internal Calabi-Yau part is described in terms of Ramond-Neveu-Schwarz variables [5]. Indeed, the hallmark of a Calabi-Yau compactification is the existence of an $N = (2, 2)$ superconformal algebra of constraints due to the

vanishing of the source of the anomaly of an axial $U(1)$ symmetry of the Σ-model in such backgrounds. On general grounds, the functions of the compactified pure spinor variables representing deformations of this background must organize into representions of this algebra. But the deformations are precisely the dynamical superfields and the representations are the primary operators of the worldsheet superconformal field theory. Thus, we recover the relation to the hybrid description of the superstring.

At present, the only understanding of the relation between the hybrid and pure spinor formalisms is by a complete breaking of the space-time super-isometries in 10 flat dimensions followed by a complicated series of field redefinitions [12]. By contrast, the only breaking of super-isometries required by the compactification itself is that reducing the structure group $\mathbb{R}^{10|16} \rtimes Spin(9,1) \to \mathbb{R}^{4|4} \rtimes Spin(3,1) \times SU(3)$. This incongruity is not merely a semantic issue. Firstly, the elucidation of the role of the pure spinor in a Calabi-Yau background holds the promise of resolving the "picture-changing problem" of the hybrid formalism in the presence of Ramond-Ramond backgrounds: While inclusion of phenomenologically interesting effects (for example, moduli stabilization, spontaneous supersymmetry breaking, space-time warping) due to Ramond-Ramond fluxes in the internal Calabi-Yau is possible order-by-order in the Σ-model loop expansion [13, 14], a problem due to the time-like chiral ρ boson of the hybrid formalism and analogous to the picture ambiguity in the RNS string requires the inclusion of new operators at each order with ever-increasing/decreasing ρ-charge. The existence of an all-orders result is difficult to ascertain, let alone obtain explicitly, although it has been done in cases with large numbers of super-isometries (most notably [15]).

Secondly, the role of the pure spinor variables in the Calabi-Yau geometry is obscured. A better tracking of the variables in the compactification would result in an understanding of the map between the geometry of the pure spinor space (defining a self-dual plane in 10-dimensional superspace) and the geometry of the Calabi-Yau manifold. For example, already in the hybrid formalism one can see the emergence [13] of the conditions on the generalized geometry [16], also called "the pure spinor equations" [17], for *a priori* different reasons. On the space-time side, the more covariant compactification would shed light on the general problem of dimensional reduction in superspace [18]. In this interpretation, it seems likely that (some components of) the pure spinor will assume the role of superspace harmonics [19, 20, 21]. Again, this expectation is partially borne out in the hybrid description [8, 22] where the role of the time-like chiral boson "ghosts" (which are related to components of the pure spinor [12]) as the auxiliary parameter of $N = 2$ projective superspace is explained. In the first case, which considers the type-II string on a K3 surface, the 2 chiral boson "ghosts" combine to form the projective parameter. In general, the pure spinor components enter the construction of the RNS fermions as well and it may not be possible to find a field redefinition to an independent set of RNS-type variables. In this case, the relevant components of the pure spinor can be expected to interact with the geometry in the internal space, related by worldsheet supersymmety from the RNS point of view. Thus we may expect an interplay between pure spinors, harmonics, and internal geometry. Indeed, in the second case the ρ-ghost of the hybrid formalism for the heterotic string on $\mathrm{K3} \times T^2$ combines with the RNS fermion of T^2 to form the auxiliary parameter of $N = 2$ projective superspace.

Holographic backgrounds Even in the maximally symmetric backgrounds much remains to be understood. Most pressing is the analysis of the amplitudes of the type-IIB pure spinor string on $\mathrm{adS}_5 \times S^5$. The two main ingredients in this recipe are the vertex operators and the superspace integration measure. It was recently shown that these have some remarkable properties in the $\mathrm{adS}_5 \times S^5$ background when compared to their flat-space analogues as we now briefly review. For

simplicity of exposition, we restrict attention to the tree-level amplitudes of massless fields.

Massless unintegrated vertex operators are easily defined as weight-0 functions on the pure spinor superspace which are linear in the pure spinor variable and in the cohomology of the BRST-like operator. To construct amplitudes, however, integrated vertex operators are needed. These are usually constructed by taking the single pole in the operator product of an unintegrated vertex operator and the b-ghost arising from the gauge fixing of worldsheet reparameterization invariance. Since, as already remarked, the pure spinor superstring is not constructed by gauge fixing a reparameterization-invariant action, there is no fundamental b-ghost in this formalism. The required integrated vertex operator is produced by constructing a weight-1 function on the pure spinor superspace which is independent of the pure spinor and has the property that the BRST-like variation of it is the worldsheet exterior derivative of the unintegrated vertex operator [23]. Alternatively, a composite b-ghost can be constructed from the first-class part of the Siegel constraints provided one is willing to break Lorentz invariance by picking a direction in spin space [24].

The 10-dimensional flat-space normalization of the 0-modes of the pure spinor variables, suppressing the ubiquitous $\int \mathrm{d}^{10}x$, is schematically given as $\int \mathrm{d}\mu\lambda^3\theta^5 = 1$ – a surprising assertion given that the pure spinor has 11 (holomorphic) degrees of freedom and the odd superspace variables carry 16. In fact, the normalization has a natural interpretation in terms of the super-symplectic formalism of Batalin-Vilkoviski where the top-degree form is related to the anti-field of the spacetime ghost. It was shown in reference [25], by a computation of the cohomology of the BRST-like operator, that a pure spinor wave function encodes the Yang-Mills gauge theory in the BV form and it was shown that the spacetime anti-ghost vertex operator at zero momentum is the unique state $\lambda^3\theta^5$. In [26] the measure was reconstructed from the superspace analogue of the construction of the Riemannian volume of a manifold using a connection on its frame bundle – a technique referred to as the "superform method" or "ectoplasm".

Considerations involving higher-loop amplitudes prompted the extension of the minimal pure spinor formalism by addition of auxiliary variables in cohomologically contractible pairs [27]. The resulting "non-minimal" pure spinor string is formally an $N = 2$ topological string and amplitudes are given by the standard prescription for such strings. In this formalism, the 0-mode integrals run over the full range of the non-minimal variables but this integral requires a regulator (which is not fully supersymmetric off-shell). The formerly Lorentz-non-invariant composite b-ghost becomes invariant in this formalism since the required choice of direction in spin space is made by the dynamical, albeit auxiliary, non-minimal variable. Thus, all apparent peculiarities of the minimal pure spinor formalism mentioned in regard to the calculation of amplitudes are explained. The price paid for this is the complication of the minimal formalism by the auxiliary variables.

Remarkably, the type-IIB superstring in $\mathrm{adS}_5 \times S^5$ background is conceptually simpler than the situation described above for flat space [28]. In this background, the Ramond-Ramond field provides a "metric" in spin space. From this Ramond-Ramond field and the right-moving pure spinor a new spinor can be constructed which, from the point of view of the target space symmetry group, behaves just as the non-minimal variable previously associated to the left-moving pure spinor. Therefore, in this background, the advantages provided by the non-minimal formalism are exploited without the need for auxiliary variables: Both the b-ghost and the regularization factor for the 0-mode integral can be constructed with just the variables present in the minimal pure spinor description of the type-IIB string and the background Ramond-Ramond flux. In fact, the situation is better yet as it turns out that the explicit form of the b-ghost so constructed is much simpler than in flat space and the regularization factor is altogether unnecessary. The resulting 0-mode measure thus reduces to the usual super-Riemannian measure over the full superspace and all components of the (projective)

pure spinors.

A further simplification is that the Lagrangian in the background is manifestly supersymmetric in the sense that it is invariant under the odd isometries (actually, it is manifestly $PSU(2,2|4)$ invariant). This is in contrast to the flat-space case in which the variation of the Lagrangian is an exact differential. This observation was extended to a statement about *all* vertex operators in the $\mathrm{adS}_5 \times S^5$ background in [29]. Operators with this type, that is, those in the $PSU(2,2|4)$-equivariant cohomology of the BRST-like operator are referred to as "universal vertices" since any state of the pure spinor string in this background can be assigned a vertex with this property in an essentially unique way.

Proposal The studies of compactified and holographic reductions of the 10-dimensional string often give rise to similar and complementary questions. During the investigations into a specific problem of the one kind, solutions to a specific problem of the other tend to suggest themselves. In addition, certain problems are inherently of both the compactification and holographic type. For example, the superspaces entering in the holographic problems on the CFT side are naturally identified with embedded submanifolds of the superspace on the adS side and can therefore be interpreted as compactifications of the latter. Conversely, in compactification problems one often studies the near-horizon geometry of BPS states which contain an adS factor and are therefore holographic. For these reasons, the proposal is to study the two branches simultaneously.

Concerning the compactification-type backgrounds, the general goal of this proposal is the understanding of compactifications of the 10-dimensional pure spinor superstrings and the exploration and exploitation of the resulting description. Specifically, a direct correspondence respecting all relevant symmetries is sought between the compactified pure spinor variables and the hybrid variables. Once the details of this correspondence have been worked out for the simplest case (*viz.* heterotic strings on manifolds of $SU(3)$ holonomy) there are various questions which can be addressed of which we name a few here.

1. What is the mechanism by which the compactified string satisfies the physical state conditions? In the 10-dimensional theory, demanding that a state be in the cohomology of the BRST-like operator was sufficient to derive the field equations due to the large amount of linearly-realized supersymmetry. In contrast, the hybrid has a full set of $N = 2$ superconformal generators which are used to define physical states in the manner familiar from the RNS string. Effectively, higher-order contractions (and, thus, higher loop order in the Σ-model) in the hybrid replace the single contractions effected by the BRST-like operator.

2. Before proceeding to the study of the type-II superstring compactifications, the implications of the existence of an $N = 2$ superconformal algebra in the heterotic string compactifications should be investigated. Here one can hope to shed some light on the (non-)existence of a non-trivial heterotic analogue of mirror symmetry. Since the existence of the worldsheet superconformal symmetry is intimately tied up with the interactions induced by the geometry of the gauge bundle and underlying Calabi-Yau manifold, one might hope to obtain concrete formulas analogous to the usual cohomological statement of mirror symmetry of Calabi-Yau manifolds.

3. How does this string avoid the picture-changing problem of the hybrid string? The results of the heterotic analysis can be applied *mutatis mutandis* to the type-II strings. These strings have minimally supersymmetric backgrounds with Ramond-Ramond flux which generate a

potential for the moduli and constitute, therefore, a phenomenologically interesting class. Does it become possible to derive all-orders equations for the string fields where in the hybrid description this required the inclusion of operators with ever-higher picture?

4. On the technical side, these strings have target superspaces with extended supersymmetry. Does (part of) the pure spinor play the role of "harmonics" in off-shell superspaces? (If the part of the pure spinor is directly related to the "ρ-ghost" of the hybrid the answer seems to be affirmative from previous study of the heterotic string on K3 $\times T^2$ [22].) In this line of investigation, the heterotic analysis could be modified to describe compactifications on $SU(2)$-holonomy manifolds leading to backgrounds with $N = 2$ supersymmetry. Elucidation of the role, if any, of the pure spinor as a harmonic/projective superspace parameter is expected in this case. The Borcea-Voisin quotient on the internal space should then be related to the natural $N = 2 \to 1$ quotient in the projective superspace target. Generalizing what is learned from this, can we construct an off-shell $N = 4$ superspace from compactification on manifolds with smaller holonomy?

With regard to the holographic-type backgrounds, the general goal is to advance our understanding of the computation of amplitudes. This requires a better understanding of the vertex operators of "universal" type and the integration over superspace.

1. With regard to the understanding of the vertex operators, the powerful machinery of homological algebra is being brought to bear [29]. This may, in fact, be necessary as the brute-force calculations so heroically carried out for the flat-space case cease to be effective. The form of the universal vertex operators in the $\mathrm{adS}_5 \times S^5$ background are superficially unusual when compared to the non-universal type familiar from the flat background. In fact, the only such vertex with non-trivial spacetime dependence currently known is constructed from a pure spinor function on superspace analogous to the pion field in the non-linear Σ-model on S^2.

2. Similarly, simplification of the superspace integration problem may require the extension of the general integration theory on Lie groups and their quotients to the category of supergroups. This general mathematical theory can be combined with the superform method for construction of the component projection formula from the superspace geometry to produce an algorithm for the important case of the $PSU(2,2|4)$ supergroup and its quotients. Besides the previously reviewed success in the application of the superform method to the flat-space case to elucidate the structure of the $\lambda^3\theta^5$ measure, a recent paper describing a shortcut of the cumbersome superform algorithm for the construction of the component density projection formula [30] may be interpreted to support this assertion. Whereas the usual procedure for the construction of the density projector takes as input a closed superform of top bosonic degree obtained by solving the Bianchi identities for it in the relevant background, this reference points out that any collection of closed superforms, the wedge of which gives a non-degenerate form of top bosonic degree, will do.[1] In principle then, one can construct the required form by wedging bi-invariant-super-1-forms. Of course, this is nothing other than the supersymmetric extension of a common method of obtaining the Haar measure on a Lie group.

[1] Actually, in this reference only the example of 4-dimensional chiral density projectors for $N = 1$ and $N = 2$ superspaces are constructed by squaring particular 2-forms but it is easy to see that the statement generalizes as claimed here.

3. Direct attack on the 10-dimensional integration problem, should it prove to be technically challenging, may be avoided by studying the mechanisms at work in lower-dimensional "toy models". These models, which have bosonic geometries of the form $\mathrm{adS}_n \times S^n \times X^{10-2n}$, where X is a $(10-2n)$-dimensional manifold with special holonomy, are of interest in their own right as they are the near-horizon geometries of extremal black holes in $2n$-dimensional spacetimes. In these lower-dimensional spaces, the construction of the vertex operators and density measures should simplify. For example, if $n = 2$ and the holonomy group of X is $SU(3)$, the target superspace is conformal to flat 4-dimensional, $N = 2$ superspace, which is well-understood. In the case where $n = 3$ and X is a K3 surface, the flat superspace is not understood as well but now the spacetime is a supergroup rather than a symmetric space.

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